RAHMAN HUSSAIN

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WORK EXPERIENCE

SUNY Research Foundation

Research Aide (ML Engineering) **Apr 2026 – May 2026**

- Built a two-stage log anomaly detector: a small fine-tuned language model screens every log line and sends only uncertain cases to a larger model.
- Fine-tuned the small model on logs from two production systems, reaching F1 scores of **0.97** and **0.90**.

Youro

Software Engineer Intern **Aug 2025 – Dec 2025**

- Designed and built a diagnosis-assist service end to end (Python, FastAPI): an LLM extracts symptoms from intake answers, then hybrid search (FAISS + BM25) pulls matching sections of urology clinical guidelines.
- It suggests the top 3 likely diagnoses with guideline citations; the urologist's diagnosis was in that list on ~**85%** of ~**60** test cases I built with him.
- Added safety checks: answers limited to the clinic's diagnosis list, every citation verified against the retrieved text, and patient identifiers removed before any API call.
- Owned the patient React and React Native apps through a beta launch with ~**20** patients, including an adaptive intake form that cut the doctor's pre-visit review from ~**15 min** to under **4 min**.

Goodz Pvt. Ltd

Software Engineer **May 2024 – Dec 2024**

Software Engineer Intern **Jul 2023 – May 2024**

- Owned the order and payments system for a marketplace of **340 vendors** (Node.js, PostgreSQL), from schema design to production support; idempotency keys and strict order states stopped duplicate orders.
- Built an automated vendor payout reconciliation pipeline against Paytm and Razorpay settlements, reducing manual finance effort from ~**6–8 hours** to ~**20 minutes** per cycle.
- Led the move from a legacy PHP site to Angular, planning a page-by-page rollout so the business never paused; page loads dropped from ~**8s** to under **2s**.
- Resolved a Paytm payment-confirmation outage within ~**30 min**, and added Redis caching that cut festival-rush menu load times from ~**6–7s** to under **200ms**.

PROJECTS

L-GRIP | IEEE S&P 2027 (in prep) | Python, PyTorch, Llama, Kafka, Neo4j **Jun 2026 – Present**

- Split anomaly detection across several small models and benchmarked it against one 70B model on the same GPU: about **4x** faster and **5x** less memory per log line.
- Built a Kafka pipeline that writes logs to a graph database and runs detection in parallel, with a check so detection never reads data before it is written.

Image Indexing & Semantic Search | UB Research | Python, FastAPI, FAISS, CLIP **Feb 2026 – Mar 2026**

- Tagged a client's photo archive with InsightFace (faces) and RAM (objects), then built search that filters in SQLite before running CLIP + FAISS vector search.

HealthAI Connect | Python, Flask, LangChain, FAISS **Aug 2025**

- Built a medical platform pairing image models that classify scans (**98.5%** test accuracy) with a RAG chatbot over ~9,000 records. Published in Taylor & Francis AI Frontiers.

SKILLS

- **AI & ML:** PyTorch, LLM fine-tuning, RAG, hybrid search (FAISS, BM25), LangChain, CLIP
- **Backend:** Python, FastAPI, Flask, Node.js, Spring Boot, REST APIs
- **Data & Infra:** PostgreSQL, Redis, Kafka, Neo4j, Docker, AWS (S3, CloudFront), Git
- **Frontend & Testing:** React, React Native, TypeScript, pytest, Jest

EDUCATION

University at Buffalo, SUNY | Master of Science in Computer Science and Engineering **May 2026**

Osmania University | Bachelor of Engineering in Information Technology **Jun 2024**